

LITTLE TREES AIR FRESHENER

ART INSTALLATION

Product Specification for Burning Man 2026

EXECUTIVE SUMMARY

A large-scale interactive art installation inspired by the iconic Little Trees car air freshener, designed to deliver a different scent experience each night of Burning Man (7-9 nights). The installation combines sculptural form, olfactory experience, lighting design, and sustainable power systems to create an immersive sensory landmark on the playa.

Estimated Total Cost: \$2,700 | Build Time: 4-6 weeks | Crew: 3-4 people

Core Requirements

- 14-ft tall cutout resembling classic Little Trees evergreen silhouette
- Different fragrance each night (manual changeover 3-5 PM daily)
- 100% solar powered with 2-3 day battery backup
- 360° LED lighting for nighttime visibility and safety
- Withstands 50+ mph wind gusts, 40-100°F temperature swings, dust storms
- Complete Leave No Trace compliance

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1. STRUCTURAL DESIGN

1.1 Dimensions

Dimension	Specification
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Project Links

GitHub Repository: <https://github.com/raywu/little-trees>

Live 3D Visualizer: <https://raywu.github.io/little-trees/>



3D Visualizer Preview

Height	14 feet (optimal for visibility)
Width at widest point	7 feet (maintains classic 1:2 proportions)
Thickness	5 inches (accommodates wiring, scent components)
Base footprint	8 x 8 feet (for stability)

1.2 Recommended Materials

Frame: Steel Tubing (Recommended)

- Material: 1.5-inch square steel tubing
- Weight: 200-300 lbs | Cost: \$640
- Pros: Superior wind resistance, durable, weldable

Skin: Corrugated Plastic (Recommended)

- Material: 6mm Coroplast sheets (4x8 ft)
- Quantity: 6 sheets | Cost: \$240
- Pros: Lightweight, translucent (LED backlighting), weather-resistant

1.3 Wind Load & Anchoring

Black Rock Desert experiences 20-30 mph sustained winds with gusts to 50+ mph. The structure must resist approximately 1,250 lbs of lateral force and 8,800 ft-lbs of overturning moment.

- Primary: 6 helical ground anchors (48-inch penetration)
- Secondary: Weighted base platform (1,000-1,500 lbs sandbags/water barrels)
- Guy wires: 6 cables at 45° angles, tensioned to 200-300 lbs each
- Safety: LED markers on all guy wires (BRC requirement)

2. SCENT DELIVERY SYSTEM

2.1 Recommended Technology: Hybrid Fan + Passive System

- Components: 3x 12V DC fans (120mm) + large felt scent pads (24x36 inches)
- Power consumption: 15-30W total
- Coverage: 30-50 ft radius (wind-dependent)
- Cost: \$230 (fans, pads, suspension system, fragrance oils)
- Pros: Best balance of effectiveness, power efficiency, reliability

2.2 Daily Scent Rotation Schedule

Night	Scent	Purpose
1	Black Ice	Classic nostalgic opening
2	Pine/Forest	Fresh, evokes the tree shape

3	Lavender	Calming contrast to playa intensity
4	Citrus	Energizing mid-week boost
5	Vanilla/Spice	Warm and comforting
6	Eucalyptus	Refreshing for dusty conditions
7-9	Sandalwood/Cedar	Spiritual closing, grounding for final days

2.3 Changeover Protocol

- Time: 3-5 PM daily (before sunset, minimal activity)
- Duration: 15-20 minutes
- Process: Turn off fans → Remove old pads → Install fresh pads with 200-500ml oil → Wait 30 min → Restart fans
- Fragrance quantity: 3-7 liters total for event | Cost: \$105

3. SOLAR POWER SYSTEM

3.1 Power Budget

Component	Watts	Hours	Wh/Day
LED lights	50	12	600
Scent fans	25	8	200
System losses	—	—	80
TOTAL	—	—	880

3.2 System Specification

- Solar panels: 2x 100W monocrystalline (200W total) | Cost: \$250
- Batteries: 2x 12V 200Ah AGM deep cycle (2,400 Wh capacity) | Cost: \$560
- Charge controller: 20A MPPT | Cost: \$180
- Wiring & protection: 10 AWG solar cables, 8 AWG battery cables, fuses | Cost: \$80
- Autonomy: 2-3 days (for dust storms)
- Daily maintenance: Clean solar panels with soft brush (critical for output)

4. LIGHTING SYSTEM

4.1 Recommended Design: LED Strip Perimeter

- Configuration: 12V LED strips around entire tree silhouette (40-50 ft total)
- Density: 60 LEDs/meter | Power: 50-75W total
- Color: Green (classic) or RGB addressable (dynamic effects)

- Runtime: Dusk to dawn (12 hours/night)
- Cost: \$120

4.2 Guy Wire Safety Lighting

- Requirement: All guy wires must be lit (BRC safety rule)
- Recommended: LED rope light wrapped around each wire OR battery LED clips every 2-3 ft
- Cost: \$40

5. BILL OF MATERIALS

Item	Qty	Unit Price	Total
Steel tubing 1.5" sq (20 ft lengths)	8	\$80	\$640
Coroplast 6mm (4'x8' sheets)	6	\$40	\$240
Ground anchors, guy wire, hardware	1 lot	—	\$290
12V DC fans (120mm), pads, frame	1 lot	—	\$125
Fragrance oils (7 scents, 8oz each)	7	\$15	\$105
Solar panels (100W monocrystalline)	2	\$125	\$250
AGM batteries (12V 200Ah)	2	\$280	\$560
MPPT charge controller (20A)	1	\$180	\$180
Wiring, fuses, connectors	1 lot	\$80	\$80
LED strip lights (12V, 50 ft total)	2 rolls	\$60	\$120
Guy wire LED markers/lights	1 lot	\$40	\$40
SUBTOTAL			\$2,480
Tools & consumables (estimate)	—	—	\$250
TOTAL PROJECT COST			\$2,730

6. INSTALLATION & OPERATION

6.1 Pre-Event Build (4-6 weeks)

- Weeks 1-2: Order materials, ensure delivery
- Weeks 2-3: Fabricate frame, weld/assemble
- Weeks 3-4: Attach skin, test-fit
- Weeks 4-5: Wire electrical, test solar/LED/scent systems
- Week 6: Final testing, pack for transport

6.2 On-Playa Installation (4-5 days)

- Day 1: Site selection, anchor installation, frame assembly (6-8 hrs)
- Day 2: Attach skin, install guy wires (4-6 hrs)
- Day 3: Solar/battery/scent system installation (4-5 hrs)
- Day 4: LED lighting, cable management, testing (3-4 hrs)
- Day 5: Final checks, first scent loading, 24hr monitoring (2 hrs)

6.3 Daily Maintenance (30-45 min/day)

- Morning: Inspect structure, clean solar panels, check battery
- Afternoon (3-5 PM): Scent changeover (15-20 min process)
- Evening: Verify lighting, check guy wires

6.4 Deinstallation & Leave No Trace (1-2 days)

- Day 1: Disconnect electrical, remove lighting, dismantle scent system
- Day 2: Remove frame, extract anchors, fill holes, MOOP sweep 50-ft radius
- Final: Document pristine site restoration with photos

7. RISK ASSESSMENT

Risk	Impact	Mitigation
High winds	Collapse/injury	Over-engineered anchoring, emergency plan
Battery depletion	Loss of safety lighting	2-3 day capacity, backup battery
Guy wire trips	Participant injury	LED lighting, reflective markers, buffer zone
Scent allergies	Respiratory irritation	Warning signage, natural oils, moderate intensity

8. CONCLUSION

This specification provides a comprehensive roadmap for creating a successful Little Trees air freshener art installation at Burning Man 2026. The design balances cost (~\$2,700), reliability (solar-powered with backup), and experiential impact (nightly scent rotation).

- Estimated budget: \$2,730 total
- Pre-event build: 4-6 weeks
- On-playa setup: 4-5 days with 3-4 person crew

- Daily maintenance: 30-45 minutes
- Expected duration: Full event with proper care

The installation prioritizes safety (anchoring, lighting), sustainability (solar power, Leave No Trace), and participant experience (olfactory surprise) while honoring the nostalgic whimsy of the Little Trees brand. With proper execution, this will serve as a memorable sensory landmark on the playa.

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Next steps: Register with Burning Man Art Department (Feb-May)